

Descriptive Qwen and NVIDIA Remote-Model Panels for Fixed-Policy LLM Play in Four-Player Go-Stop

Anonymous

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Abstract

We evaluate fixed-policy large language model play in four-player Go-Stop, a Korean card game with stochastic deals, seat-order nuisance effects, multi-agent interaction, and risk-sensitive payoffs. The study asks two constrained empirical questions: whether observed performance across a Qwen-labeled remote-model panel is monotone in nominal parameter scale, and whether cross-family differences are visible in a rate-limited, roughly same-scale NVIDIA Build panel. We run two remote-API panels with exhaustive seat permutations and repeated layouts. The Qwen panel compares four DashScope models spanning 30B, 122B, 397B, and 480B total parameters. The NVIDIA Build panel compares four roughly 100B–122B models from Qwen, Mistral, Nemotron, and Stockmark. Each panel yields 48 session samples per model and 192 model-session observations. The Qwen descriptive ranking is non-monotonic: `qwen3.5-397b-a17b` obtains the highest mean profit and least severe empirical $\text{CVaR}_{0.05}$, while the larger `qwen3-coder-480b-a35b-instruct` ranks lower. In the constrained NVIDIA panel, `nvidia/nemotron-3-super-120b-a12b` has the highest mean profit and least severe empirical $\text{CVaR}_{0.05}$. These results are descriptive, not universal model rankings: they show that this harness can expose policy differences that are not summarized by nominal parameter count alone. We release the evaluation harness, manifests, logs, and result tables as reproducible artifacts.

1 Introduction

This paper is about evaluation, not training. We study fixed language-model policies in four-player Go-Stop, where agents must choose among legal game actions under partial information, stochastic deals, seat-order effects, and heavy-tailed settlements. The goal is not to build a strong Go-Stop player, but to ask whether remote LLM policies exhibit measurable differences under a reproducible game harness.

The motivating claim is simple: model scale is an incomplete proxy for game-playing policy quality. Larger models often have stronger general capabilities, but a multi-agent, risk-sensitive game can expose differences in action selection, risk tolerance, and instruction following that are not captured by parameter count alone. Conversely, comparing unrelated model families can confound architecture, training mixture, alignment, tool policy, and provider behavior. We therefore report two bounded descriptive panels rather than a universal ranking.

- **RQ1:** In a Qwen-labeled DashScope panel, is observed Go-Stop performance monotone in nominal parameter scale?
- **RQ2:** In a rate-limited NVIDIA Build panel of roughly same-scale models, are cross-family differences visible in descriptive profit and risk summaries?

The answer to RQ1 is descriptively negative in this panel: the 397B Qwen3.5 model ranks ahead of the 480B Qwen Coder model on mean profit and empirical $\text{CVaR}_{0.05}$. This comparison is confounded by model variant and specialization, so it should not be read as an isolated estimate of scale. For RQ2, the constrained NVIDIA panel descriptively ranks Nemotron highest on mean profit and empirical tail risk. These are panel-specific findings under a fixed harness, fixed prompt, and explicit remote-API constraints.

2 Related Work

Game environments are useful because they make policy differences observable under controlled stochasticity. OpenSpiel is the canonical reference for game evaluation infrastructure [Lanctot et al., 2019, DeepMind, 2019]. Our setting is narrower and more applied: we implement a Go-Stop harness rather than a general game suite, and our agents are frozen remote LLM policies rather than trained reinforcement-learning policies.

The multi-agent aspect connects to zero-shot coordination and ad hoc teamplay work, where fixed policy diversity and behavioral mismatch become the object of study [Hu et al., 2020, Lucas and Allen, 2022, Jeon and Kim, 2023, Nekoei et al., 2023]. The risk-sensitive aspect matters because Go-Stop settlements can be asymmetric and heavy-tailed. We therefore report $\text{CVaR}_{0.05}$ and ruin probability alongside mean profit [Eriksson et al., 2022, Wang et al., 2023]. For finite-sample summaries, the repository uses explicit manifests, sample counts, empirical tail-support fields, and inferential-readiness diagnostics rather than hiding uncertainty behind point estimates.

3 Methods

3.1 Environment

The harness evaluates fixed policies only. There is no online learning, self-play training, or log-based policy update. The implemented game starts with four players and proceeds to a three-active-player main game after participation decisions. The engine includes jokers, same-month match choices, scoring categories, Go/Stop decisions, voluntary exit, forced gwang sell, showdown proposals, and carryover multipliers. Each agent receives a public game-state serialization and must return one legal action index.

3.2 Policy Interface

Remote models are called through OpenAI-compatible chat-completion APIs. The policy prompt is fixed:

```
You are a fixed Go-Stop evaluation policy. Choose one legal action index only.
Optimize long-run profit with risk control. Do not explain.
```

At each decision, the model is instructed to return JSON with a single `choice_index`. Invalid, unparsable, or out-of-range responses fall back to the first legal action. Repeated public states are cached by hash.

3.3 Panel A: Qwen Parameter-Scale Study

The Qwen study uses Alibaba DashScope and compares four Qwen-labeled models:

- qwen3-coder-30b-a3b-instruct

- qwen3.5-122b-a10b
- qwen3.5-397b-a17b
- qwen3-coder-480b-a35b-instruct

The total parameter counts are 30B, 122B, 397B, and 480B. The active parameter counts recorded in the manifest are 3B, 10B, 17B, and 35B. Because the panel mixes Qwen3.5 and Qwen Coder variants, it tests monotonicity in this callable Qwen-labeled panel rather than an isolated scale effect. We enumerate all $4! = 24$ seat permutations and repeat each layout twice, producing 48 session samples per model and 192 model-session observations. Each session has a two-hand horizon. Decoding is fixed at `temperature=0`, `max_tokens=64`, and `enable_thinking=false`.

3.4 Panel B: NVIDIA Constrained Same-Scale Family Study

The NVIDIA study uses NVIDIA Build NIM and compares four roughly 100B–122B models:

- qwen/qwen3.5-122b-a10b
- mistralai/mistral-small-4-119b-2603
- nvidia/nemotron-3-super-120b-a12b
- stockmark/stockmark-2-100b-instruct

The recorded parameter counts are 122B, 119B, 120B, and 100B. We again enumerate all 24 seat permutations and repeat each layout twice, producing 48 session samples per model and 192 model-session observations. Because NVIDIA rate limits were binding in live execution, this panel uses a one-hand horizon and `max_remote_calls_per_agent=1`; after each agent’s one remote call budget is exhausted, the harness uses the same first-legal-action fallback used for other remote-call limits. This makes the NVIDIA study a constrained first-decision proxy rather than a full-horizon autonomous-play comparison.

3.5 Metrics

The primary metric is terminal session profit relative to the initial bankroll. We also report empirical $\text{CVaR}_{0.05}$, win rate, ruin probability, and sample count. CVaR is computed over session profits. Each model has 48 session samples; the effective 5% tail mass is therefore 2.4 sessions and the empirical support count is 3. The artifact table records `cvar_5_low_support`; this flag is false for all rows because the implementation marks support counts below 3 as low support. The statistical report separately marks CVaR inference below 100 sessions per model as unstable, so CVaR is treated here as a descriptive tail summary.

4 Results

4.1 RQ1: Qwen Parameter Scale

Table 1 summarizes the Qwen panel. The highest descriptive mean profit and least severe empirical $\text{CVaR}_{0.05}$ belong to `qwen3.5-397b-a17b`. It achieves mean profit 996.25 and $\text{CVaR}_{0.05}$ of -1850.08. The 480B Qwen Coder model ranks lower: it has mean profit -198.92 and $\text{CVaR}_{0.05}$ of -5084.75. The 122B model has the highest win rate, 0.3542, but its mean profit and empirical tail summary are lower than the 397B model.

Table 1: Qwen parameter-scale panel. Each model has 48 session samples. The CVaR low-support flag is false for all rows, but the effective 5% tail mass is 2.4 sessions, so CVaR is descriptive.

Model	Total params (B)	Active params (B)	Mean profit	CVaR _{0.05}	Win rate
qwen3.5-397b-a17b	397	17	996.25	-1850.08	0.3021
qwen3.5-122b-a10b	122	10	-51.75	-4145.33	0.3542
qwen3-coder-480b-a35b-instruct	480	35	-198.92	-5084.75	0.2396
qwen3-coder-30b-a3b-instruct	30	3	-745.58	-5269.25	0.1042

Table 2: NVIDIA constrained roughly same-scale family panel. Each model has 48 session samples. The CVaR low-support flag is false for all rows, but the effective 5% tail mass is 2.4 sessions. The run uses `max_remote_calls_per_agent=1`.

Model	Params (B)	Mean profit	CVaR _{0.05}	Win rate
nvidia/nemotron-3-super-120b-a12b	120	500.65	-1384.17	0.2708
mistralai/mistral-small-4-119b-2603	119	-73.19	-4150.00	0.2917
qwen/qwen3.5-122b-a10b	122	-172.85	-3350.42	0.1875
stockmark/stockmark-2-100b-instruct	100	-254.60	-4267.08	0.2500

The observed ranking is non-monotonic. A larger nominal parameter count is not sufficient to order performance in this panel. This is clearest in the contrast between the 397B Qwen3.5 model and the 480B Qwen Coder model. Because that contrast also changes model variant and specialization, the panel does not isolate a pure scale effect.

4.2 RQ2: NVIDIA Constrained Family Panel

Table 2 summarizes the NVIDIA panel. The highest descriptive mean profit and least severe empirical CVaR_{0.05} belong to `nvidia/nemotron-3-super-120b-a12b`, with mean profit 500.65 and CVaR_{0.05} of -1384.17. Mistral has the highest win rate, 0.2917, but its mean profit and empirical tail summary rank below Nemotron’s. Qwen and Stockmark are both negative on mean profit in this constrained panel.

This is consistent with cross-family differences being visible under the run constraints, but it is not a confirmatory family-effect estimate. The compared models are close in nominal scale, but the run also includes provider behavior, model specialization, and a one-call request budget. Nemotron combines positive mean profit with the least severe empirical lower tail in this constrained panel.

5 Discussion

The two panels agree on a bounded point: fixed-policy Go-Stop performance in this harness is not fully summarized by nominal parameter count. In the Qwen panel, the highest descriptive result comes from the 397B model rather than the 480B model. In the NVIDIA panel, roughly same-scale models separate in descriptive profit and empirical tail summaries, with Nemotron ranking first.

These outcomes fit the game domain. Go-Stop requires local tactical choices, risk control, settlement awareness, and disciplined JSON action selection. General language-model scale can help, but the observed policy is also shaped by model specialization, instruction following, decoding behavior, provider-level serving constraints, and fallback behavior. The 480B Qwen Coder result is a warning against treating larger or more code-specialized models as automatically stronger game

policies under a compact action-index prompt.

6 Limitations

The results should be stated with explicit boundaries.

- The model panels are convenience samples from currently callable remote APIs.
- The Qwen scale panel mixes Qwen3.5 and Qwen Coder variants, so parameter scale is confounded with model specialization.
- The NVIDIA panel is constrained by rate limits and uses `max_remote_calls_per_agent=1`; it measures constrained first-decision-policy behavior more than full-game autonomous play.
- The horizons are short: two hands for Qwen and one hand for NVIDIA.
- Each model has 48 session samples, below the repository’s 100-session threshold for stable CVaR inference.
- The exported tables do not report invalid-response, API-error, cache-hit, or fallback-action rates, although fallback behavior can affect observed policy quality.
- The sample count is enough to compute descriptive summaries but not enough for universal model ranking claims.

7 Conclusion

In a reproducible four-player Go-Stop evaluation harness, the observed Qwen panel is not monotone in nominal parameter scale, and the constrained NVIDIA Build panel shows visible cross-family differences in descriptive profit and empirical tail summaries. The highest Qwen result comes from `qwen3.5-397b-a17b`; the highest NVIDIA result comes from `nvidia/nemotron-3-super-120b-a12b`. The main implication is methodological: multi-agent, risk-sensitive card-game evaluation should report model identity, parameter scale, family, serving constraints, fallback policy, and tail-risk metrics together rather than reducing model quality to size alone.

Code and Artifact Availability

The manuscript source is under `paper/`. The main result bundles are:

- Qwen: `results/paper_qwen_4model_param_2h_2rep`
- NVIDIA: `results/paper_nvidia_120b_4model_family_1h_2rep_budget1`

Each directory includes `manifest.json`, `report.json`, `agent_performance_table.csv`, `session_observations.csv`, `session_logs.jsonl`, and `cross_play_results.json`. A compact result summary is provided in `paper/remote_model_results.md`.

References

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A Reproducibility Commands

The Qwen panel was run with:

```
python3 scripts/run_dashscope_qwen_parameter_sweep.py \  
  --session-hands 2 \  
  --layout-repetitions 2 \  
  --remote-eval-hands 2 \  
  --decoding max_tokens=64 \  
  --decoding temperature=0 \  
  --decoding enable_thinking=false \  
  --output-dir results/paper_qwen_4model_param_2h_2rep
```

The NVIDIA panel was run with:

```
python3 scripts/run_nvidia_same_size_family_eval.py \  
  --session-hands 1 \  
  --layout-repetitions 2 \  
  --remote-eval-hands 1 \  
  --max-remote-calls-per-agent 1 \  
  --decoding max_tokens=64 \  
  --decoding temperature=0 \  
  --output-dir results/paper_nvidia_120b_4model_family_1h_2rep_budget1
```

B Artifact Integrity

Both completed result directories record SHA-256 hashes for the main exported artifacts in `manifest.json`. The Qwen run has `lineup_count=48` and `session_observation_count=192`. The NVIDIA run has `lineup_count=48` and `session_observation_count=192`.